Project Plan : Synthetic Data for Entity Linking Domain Adaptation

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Short Description

Entity Linking (EL) is the task of mapping ambiguous textual mentions to their corresponding entities in a Knowledge Base (KB) or Knowledge Graph (KG). Existing approaches are mostly tied to general-domain resources such as Wikipedia/Wikidata, and adapting them to specialized domains (e.g., law, biomedicine) is hindered by the scarcity of annotated in-domain data.

This project aims to design a synthetic data generation pipeline tailored to a target domain and KG, validate it by fine-tuning models on the generated data, and benchmark the results against zero-shot transfer baselines and models fine-tuned on real human-annotated data.

The pipeline follows two main stages: (1) candidate generation using BM25 or dense retrieval (embedding model + FAISS index), and (2) entity selection via a multiple-choice QA approach with an LLM, or alternatively a reranker-based approach scoring all candidates and returning the top-1.

Hypothèses

- Synthetic data generated from a KG can effectively substitute scarce human-annotated in-domain data for fine-tuning entity linking models.
- Fine-tuned models on synthetic data will outperform zero-shot baselines on domain-specific entity linking benchmarks.

To do list

Task	Description	Person	Deadline
Setting up the work space	create a git repository make a project plan	Soline	30/04
Subgraph extraction	Choose a target domain and extract the subgraph of this domain from wikidata. Write a script to query wikidata and keep only relevant triplets.	Soline	08/05

Generate candidates	Implement BM25 on the entity from the subgraph. Write a function that takes a mention and returns the N most similar entities. Do the same with Qwen3-Embedding-4B and FAISS.	Soline	15/05
Synthetic data generation	Write a prompt asking the LLM to generate a text that naturally mentions the entities from a KG (Wikidata or a subgraph of Wikidata) Test it manually on a few examples via the OpenAI API Write a script that calls the API in a loop over all triples from the subgraph Automatically annotate the generated texts.	Luciano and Nour	29/05
Fine tune	Prepare the data for TLR and Unsloth Train on a test set to be sure that the algorithm runs correctly Train on the whole cluster	Anas	5/06
Zero shot	We take Qwen3-4B-Instruct as is, we send it our prompts, and we look at what it answers.	Lobna	15/05
Fine tune shot	We take Qwen3-4B-Instruct fine tuned, we send it our prompts, and we look at what it answers.	Anas	12/06
Benchmarking	We compare the zero shot, the fine tune on synthetic data and the fine tune on real data.	Lobna	24/06
Write the final report		All the team	26/06

Data

Wikidata is used as the starting Knowledge Graph. We will extract domain-specific subgraphs (e.g. medicine, law) by querying relevant entities and triples. These subgraphs will serve both as the target KB for entity linking and as the source for synthetic data generation.

ELGOLD (2025) is the benchmark used for final evaluation. It contains manually annotated entity linking data across multiple domains, allowing us to compare our approach against gold standard annotations.

Synthetic data will be generated by prompting a large LLM via API to produce texts that naturally mention entities from KG triples. These texts will then be automatically annotated and used to fine-tune Qwen3-4B-Instruct.

Models

BM25 is used as a keyword-based candidate retrieval method. Given a mention, it searches for the most lexically similar entities in the subgraph based on word frequency and rarity. It serves as a simple and fast baseline for candidate generation.

Qwen3-4B-Instruct is the main LLM used for entity selection. It receives the mention and a list of candidates formatted as a multiple-choice question and returns the index of the correct entity. This model will be evaluated in a zero-shot setting and then fine-tuned on our synthetic.

Risks and mitigation

Synthetic data of insufficient quality	Iterate on the generation prompt and add filtering steps to remove low quality examples
Compute constraints	Use the OpenAI API for data generation and use smaller models for fine-tuning
Wikidata subgraph extraction is too noisy	Apply strict filtering on entity types and relations relevant to the target domain

Planning week by week

Week 1 (27/04 to 3/05) : Setting up the workspace

Week 2 (4/05 to 10/05) : Subgraph extraction, Start synthetic data generation, start zero shot, start fine tune.

Week 3 (11/05 to 17/05) : Generate candidates, continue synthetic data generation, finish zero shot, continue fine tune.

Week 4 (18/05 to 24/05) : Continue synthetic data generation, continue fine tune.

Week 5 (25/05 to 31/05) : Finish synthetic data generation

Week 6 (1/06 to 7/06) : Start the fine tune shot

Week 7 (8/06 to 14/06) : Finish the fine tune shot

Week 8 (15/06 to 21/06) : Start the benchmark

Week 9 (22/06 to 26/06) : Finish benchmark and write the report